

Lists and indexing

Project: Python readiness portfolio

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Distinguish a list index from the value stored at that position

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Loops and trace tables

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each loop iteration

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

Look for: Translate stated requirements into comparisons

Reference example, not the only acceptable wording: (minutes $\leq$ 12) and quiet. At most includes equality, so $\leq$ is required; the nonnegative integer and Boolean input contract still applies.

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24. Assignment alone produces no displayed output in this script.

Teach lists as ordered collections, with index and value kept distinct. The first exercises use nonnegative indices, although Python also permits negative indexing. Reference: <https://docs.python.org/3/tutorial/introduction.html#lists>. Do not imply that updating a record changes or verifies the physical object represented.

Outcomes

Distinguish a list index from the value stored at that position

Read and update an item using a valid index

Trace a loop through list values and test an empty list

Preparation

Python 3 editor and lists.py

Four position cards numbered 0 through 3 and movable value cards

Blank trace table and separate predicted and observed output columns

Prerequisite answer

0, 1 and 2; three iterations. The stop value 3 is excluded.

End goal for the pictorial case

Trace lists and indexing and explain the changed program before running it.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

A fictional supply shelf has three trays with counts 4, 7 and 4 in that order. Store them as `counts = [4, 7, 4]`. Square brackets enclose the list and commas separate its items. The order matters: the first and third trays contain equal values but occupy different positions.

Python counts positions from zero. `counts[0]` reads the first item, `counts[1]` the second and `counts[2]` the third. `len(counts)` is 3 because there are three items. It is a count, not the last nonnegative index. This lesson uses integer counts and nonnegative indices; Python also supports negative indices, which are outside our current indexing convention.

The printed lines are 3, 4, 7 and `[4, 9, 4]`. The assignment `counts[1] = 9` replaces the item at index 1; it does not add a fourth item or change the two other values. The length stays 3. A recorded count can be changed by code without proving that the physical tray really contains that many objects.

Term · Meaning

Term: List

Meaning: An ordered collection of items

Term: Index

Meaning: An integer position used to access an item

Term: Length

Meaning: The number of items in a collection

Term: Item

Meaning: One value stored in a collection

Term: IndexError

Meaning: An error raised when an index is outside a sequence

Positions and stored counts

Tray position · Index · Value

Tray position: First

Index: 0

Value: 4

Tray position: Second

Index: 1

Value: 7

Tray position: Third

Index: 2

Value: 4

Equal stored values do not imply equal positions. There is no item at index 3.

For [2, 0, 5], totals become 2, 2, 7. Output is 7. Zero is still an item and its iteration still executes. Exit: len([8, 3]) is 2, but index 2 is outside the list. Last nonnegative index is 1, whose value is 3. An empty list has no last item, so a length check is required before access.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

```
counts = [4, 7, 4]

total = 0

for count in counts:
    total = total + count
```

```
print (total)
```

4 · 0 · 4

4: 7

0: 4

4: 11

4: 4

0: 11

4: 15

This fresh script prints 15. for count in counts supplies the stored values, not the indices. Both copies of 4 are processed because both are items. For an empty list written [], the body runs zero times and total stays 0. For [7], it runs once and prints 7.

Change only the list to [2, 0, 5]. Trace the three additions and state the final output. Explain why the middle item is processed even though its value is zero.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Lists and indexing

Indexes are positions, not values. A list of shelf counts is read by position.

The facts we will use

Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).

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Zero-based list positions

	Index 0	Index 1	Index 2
Stored counts	6	1	8

Read a cell using both its row and column labels.

Lists and indexing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Draw positions

Indexes 0,1,2 contain 6,1,8 respectively.



2. Read two positions

Index 0 returns 6; index -1 returns the final value 8.



3. Check length boundary

Length is 3, but index 3 is beyond the final valid positive index 2.

CHANGE ONE THING

Replace the second print with `print(counts[3])`.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Draw positions

What happens and why: Indexes 0,1,2 contain 6,1,8 respectively.

Step: 2. Read two positions

What happens and why: Index 0 returns 6; index -1 returns the final value 8.

Step: 3. Check length boundary

What happens and why: Length is 3, but index 3 is beyond the final valid positive index 2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

List positions begin at zero. A negative index counts from the end; the list length counts entries and is not itself a valid positive index for the final entry.

What this example does not establish

A valid position does not establish that the stored count is accurate; empty lists also have no final element to read.

Change one thing in the pictured case

Change: Replace the second print with `print(counts[3])`.

Prediction: The first print shows 6, then `IndexError` stops the script before the length print.

Reason: The attempted position does not exist.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Retrieve `range(3)` and its excluded stop.

Time or stage: 7 to 20 minutes

Teacher action: Place value cards under position cards and complete W1.

Time or stage: 20 to 33 minutes

Teacher action: Run the worked read and update script, then complete W2.

Time or stage: 33 to 45 minutes

Teacher action: Trace the value loop and answer the guided question.

Time or stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between length and final index.

Time or stage: 8 to 20 minutes

Teacher action: Predict the faulty access before running; use W3 to repair value versus index confusion.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently, keeping the replacement step separate from empty-list summation.

Time or stage: 35 to 45 minutes

Teacher action: Discuss W5 and use the exit to diagnose indexing assumptions.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Keep position cards fixed while moving value cards. Permit an oral trace and provide a blank grid.

Extension: write a guarded final-item reader that prints Empty when `len(readings) == 0`, otherwise prints `readings[len(readings) - 1]`. Test `[]` and `[6]`.

Reduce help deliberately

Completion task: Start with `counts = [4, 7, 4]`. Run `counts[0] = 6`. Write the resulting list, its length and the value at index 2. Predict before running.

Teacher check: The result is `[6, 7, 4]`; length remains 3; index 2 still holds 4. Only the first item was replaced.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Lists and indexing. Proposed explanation: “Length is the last nonnegative index”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Length counts items; the last nonnegative index of a nonempty list is one less.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Label the three index positions. Predict all original outputs, then name the first output and stopping point after using index 3.

Reasoned teacher answer

Original outputs are 6,8,3. After the change, 6 prints first, counts[3] raises IndexError, and the third print does not execute.

Changed-case reasoning: The first print shows 6, then IndexError stops the script before the length print. The attempted position does not exist.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

Length is the last nonnegative index: Length counts items; the last nonnegative index of a nonempty list is one less.

A loop variable always contains an index: A direct list loop supplies stored values.

Updating one item grows the list: Assignment at an existing index replaces that item; length is unchanged.

Repeated values are processed only once: Each position contributes one item, even when values repeat.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

For $[2, 0, 5]$, totals become 2, 2, 7. Output is 7. Zero is still an item and its iteration still executes. Exit: $\text{len}([8, 3])$ is 2, but index 2 is outside the list. Last nonnegative index is 1, whose value is 3. An empty list has no last item, so a length check is required before access.

W1 Separate index and value

For $\text{counts} = [4, 7, 4]$, give the length, valid nonnegative indices and value at index 1. Explain how the two items with value 4 differ.

Length 3; indices 0, 1, 2; $\text{counts}[1]$ is 7. The equal values occupy different positions, indices 0 and 2.

W2 Trace an update

Start with $\text{counts} = [4, 7, 4]$. Run $\text{counts}[0] = 6$. Write the resulting list, its length and the value at index 2. Predict before running.

The result is $[6, 7, 4]$; length remains 3; index 2 still holds 4. Only the first item was replaced.

W3 Repair a position mistake

A learner loops with `for count in counts` and tries to read `counts[count]` for `counts = [4, 7, 4]`. Identify the first attempted index and error. Explain how to print each value directly.

The first count is 4, so `counts[4]` raises `IndexError`. The loop variable already holds each value. Use an indented `print(count)` to print 4, 7 and 4 on separate lines.

W4 Process new data independently

Start with `readings = [3, 0, 3, 2]`. Replace the second item with 5 using its index. Write a loop that adds all updated readings and prints one final total. Trace every update of the total. Then test the summing part alone on `[]` and `[6]`.

Use `readings[1] = 5`, giving `[3, 5, 3, 2]`. Initialize `total = 0` before for reading in readings, add reading inside, and print outside. Running totals are 3, 8, 11, 13; final 13. Summing `[]` gives 0 and `[6]` gives 6. The replacement line itself requires at least two items.

W5 Keep the information you need

Compare `[1, 5, 2]` and `[5, 1, 2]`. State their totals and first items. Would storing only the total answer which first item was larger? Explain.

Both total 8. Their first items are 1 and 5, so the second list has the larger first item. A total alone discards the order and cannot answer this question.

Independent assessment and fresh retry

W4 must use index 1 for the second item, retain all four items, produce the full trace and output 13, and obtain 0 and 6 when testing only summation. After support, retry `[2, 2, 1]`, replace the first item with 4, and sum: updated list `[4, 2, 1]`, totals 4, 6, 7. Then sum `[]` and `[5]` for outputs 0 and 5. Record support before judging independence.

Assess independent transfer

Selected case: Predict the three outputs. If the second print uses `values[3]` instead, where does execution stop and which later output is not produced?

```
values = [4, 9, 2]
print(values[1])
print(values[-1])
print(len(values))
```

Selected case answer

Outputs are 9,2,3. With `values[3]`, the first print produces 9, then `IndexError` occurs; the final length print is not reached. Positive valid indices are 0,1,2.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled lists and indexing evidence sheet to Python readiness portfolio. Show how you can distinguish a list index from the value stored at that position. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Keep position cards fixed while moving value cards. Permit an oral trace and provide a blank grid. Extension: write a guarded final-item reader that prints Empty when `len(readings) == 0`, otherwise prints `readings[len(readings) - 1]`. Test `[]` and `[6]`.

Week 8 groups reusable instructions in functions and supplies values through arguments.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: python-bridge/8, python-bridge/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

7. Prepare, teach and assess

Teaching target: Distinguish a list index from the value stored at that position

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Start with `counts = [4, 7, 4]`. Run `counts[0] = 6`. Write the resulting list, its length and the value at index 2. Predict before running.

Reasoned practice answer: The result is `[6, 7, 4]`; length remains 3; index 2 still holds 4. Only the first item was replaced.

Check every learner before discussion: Use the worked case for Lists and indexing. Proposed explanation: “Length is the last nonnegative index”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Length counts items; the last nonnegative index of a nonempty list is one less.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace the second print with `print(counts[3])`. Require a prediction and a reason before comparing. Expected result: The first print shows 6, then `IndexError` stops the script before the length print. Reason: The attempted position does not exist.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.